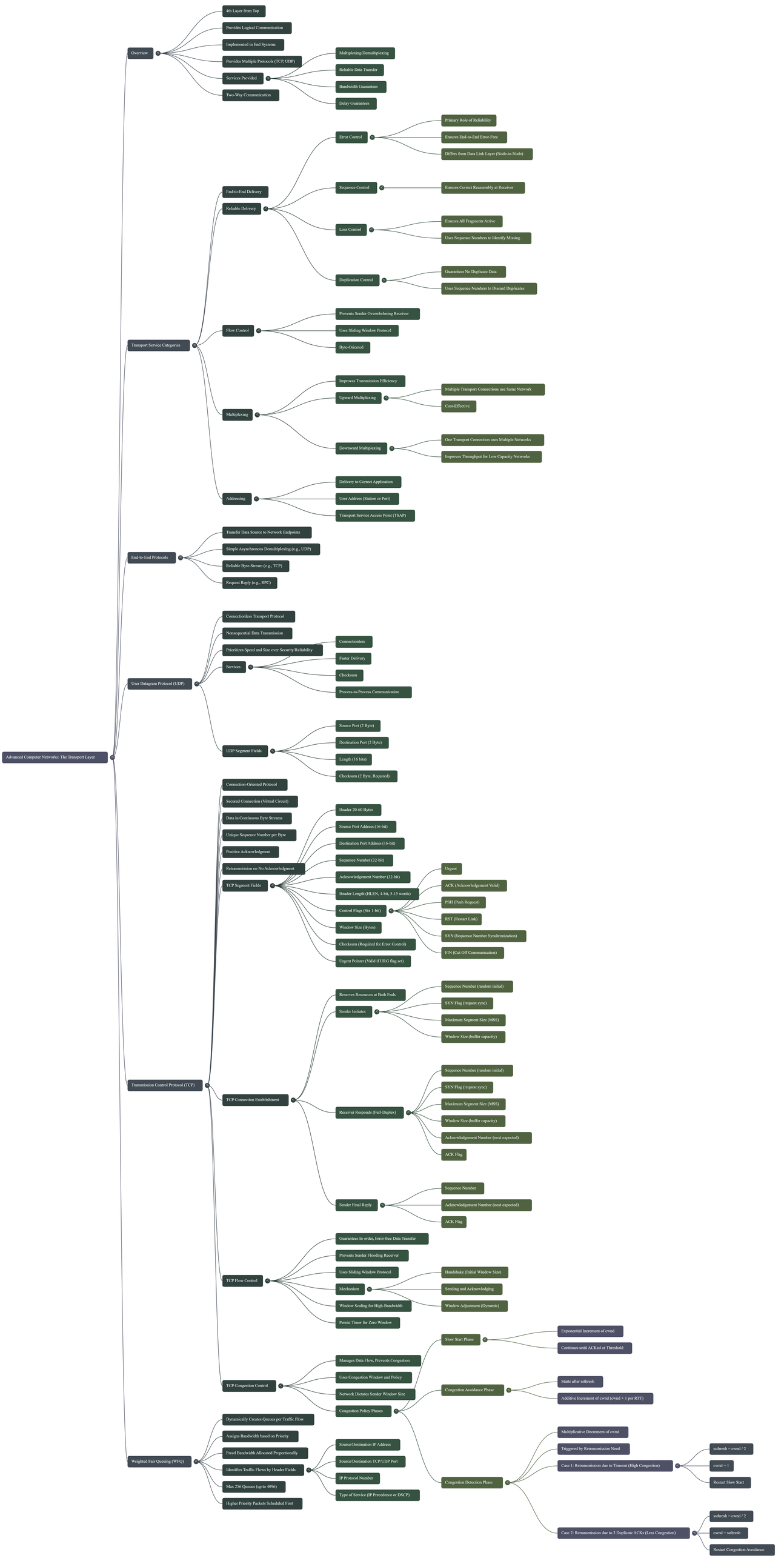
Advanced Computer Networks: The Transport Layer
Overview
4th Layer from Top
Provides Logical Communication
Implemented in End Systems
Provides Multiple Protocols (TCP, UDP)
Services Provided
Multiplexing/Demultiplexing
Reliable Data Transfer
Bandwidth Guarantees
Delay Guarantees
Two-Way Communication
Transport Service Categories
End-to-End Delivery
Reliable Delivery
Error Control
Primary Role of Reliability
Ensures End-to-End Error-Free
Differs from Data Link Layer (Node-to-Node)
Sequence Control
Ensures Correct Reassembly at Receiver
Loss Control
Ensures All Fragments Arrive
Uses Sequence Numbers to Identify Missing
Duplication Control
Guarantees No Duplicate Data
Uses Sequence Numbers to Discard Duplicates
Flow Control
Prevents Sender Overwhelming Receiver
Uses Sliding Window Protocol
Byte-Oriented
Multiplexing
Improves Transmission Efficiency
Upward Multiplexing
Multiple Transport Connections use Same Network
Cost-Effective
Downward Multiplexing
One Transport Connection uses Multiple Networks
Improves Throughput for Low Capacity Networks
Addressing
Delivery to Correct Application
User Address (Station or Port)
Transport Service Access Point (TSAP)
End-to-End Protocols
Transfer Data Source to Network Endpoints
Simple Asynchronous Demultiplexing (e.g., UDP)
Reliable Byte-Stream (e.g., TCP)
Request Reply (e.g., RPC)
User Datagram Protocol (UDP)
Connectionless Transport Protocol
Nonsequential Data Transmission
Prioritizes Speed and Size over Security/Reliability
Services
Connectionless
Faster Delivery
Checksum
Process-to-Process Communication
UDP Segment Fields
Source Port (2 Byte)
Destination Port (2 Byte)
Length (16 bits)
Checksum (2 Byte, Required)
Transmission Control Protocol (TCP)
Connection-Oriented Protocol
Secured Connection (Virtual Circuit)
Data in Continuous Byte Streams
Unique Sequence Number per Byte
Positive Acknowledgment
Retransmission on No Acknowledgment
TCP Segment Fields
Header 20-60 Bytes
Source Port Address (16-bit)
Destination Port Address (16-bit)
Sequence Number (32-bit)
Acknowledgement Number (32-bit)
Header Length (HLEN, 4-bit, 5-15 words)
Control Flags (Six 1-bit)
Urgent
ACK (Acknowledgement Valid)
PSH (Push Request)
RST (Restart Link)
SYN (Sequence Number Synchronization)
FIN (Cut Off Communication)
Window Size (Bytes)
Checksum (Required for Error Control)
Urgent Pointer (Valid if URG flag set)
TCP Connection Establishment
Reserves Resources at Both Ends
Sender Initiates
Sequence Number (random initial)
SYN Flag (request sync)
Maximum Segment Size (MSS)
Window Size (buffer capacity)
Receiver Responds (Full-Duplex)
Sequence Number (random initial)
SYN Flag (request sync)
Maximum Segment Size (MSS)
Window Size (buffer capacity)
Acknowledgement Number (next expected)
ACK Flag
Sender Final Reply
Sequence Number
Acknowledgement Number (next expected)
ACK Flag
TCP Flow Control
Guarantees In-order, Error-free Data Transfer
Prevents Sender Flooding Receiver
Uses Sliding Window Protocol
Mechanism
Handshake (Initial Window Size)
Sending and Acknowledging
Window Adjustment (Dynamic)
Window Scaling for High-Bandwidth
Persist Timer for Zero Window
TCP Congestion Control
Manages Data Flow, Prevents Congestion
Uses Congestion Window and Policy
Network Dictates Sender Window Size
Congestion Policy Phases
Slow Start Phase
Exponential Increment of cwnd
Continues until ACKed or Threshold
Congestion Avoidance Phase
Starts after ssthresh
Additive Increment of cwnd (cwnd + 1 per RTT)
Congestion Detection Phase
Multiplicative Decrement of cwnd
Triggered by Retransmission Need
Case 1: Retransmission due to Timeout (High Congestion)
ssthresh = cwnd / 2
cwnd = 1
Restart Slow Start
Case 2: Retransmission due to 3 Duplicate ACKs (Less Congestion)
ssthresh = cwnd / 2
cwnd = ssthresh
Restart Congestion Avoidance
Weighted Fair Queuing (WFQ)
Dynamically Creates Queues per Traffic Flow
Assigns Bandwidth based on Priority
Freed Bandwidth Allocated Proportionally
Identifies Traffic Flows by Header Fields
Source/Destination IP Address
Source/Destination TCP/UDP Port
IP Protocol Number
Type of Service (IP Precedence or DSCP)
Max 256 Queues (up to 4096)
Higher Priority Packets Scheduled First